Statistics and Data Analysis for Engineers

A Practical, Python-Oriented Introduction



Richard Brooks

VIA University College

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Part I

Data and Probability Models

Chapter 1 Data and Descriptive Statistics

Engineering decisions often begin with a simple question: what does the data actually say? A production engineer may have hundreds of thickness measurements, a software engineer may have thousands of response times, and a test engineer may have counts of several defect types. Raw observations are rarely useful on their own. They must be organised, summarised, visualised, and interpreted in their engineering context.

This chapter develops that first analysis. Its definitions and notation follow the descriptive-statistics treatment in *Mathematics for Software Engineering*, while the workflow is adapted to STA1: we begin with an engineering dataset, use Python throughout, and separate descriptive evidence from inferential conclusions. Formal probability models and statistical inference come later.

1.1 Engineering question and learning outcomes

Our running example concerns the thickness of manufactured plates. A nominal value near $250\ \mu$m is expected, and measurements have been collected from two production lines. We want to answer:

> How can the measurements be summarised without hiding important variation, unusual observations, or differences in the data-collection process?

After studying this chapter, you should be able to:

- distinguish populations, samples, parameters, and statistics;
- classify variables and choose appropriate summaries and plots;
- compute and interpret measures of centre, position, and spread;
- explain why sample variance uses the divisor $n - 1$;
- describe shape using symmetry, skewness, and modality;
- use the $1.5 \times IQR$ rule to flag observations for investigation;
- create and interpret frequency tables, histograms, boxplots, and scatterplots in Python; and
- state an engineering conclusion that respects the limits of descriptive analysis.

1.2 Populations, samples, and variables

Definition 1.1 (Population)

A **population** is the complete set of observations or units relevant to a stated question. ♣

The population is determined by the question, not merely by what happens to be in a file. For the plate example, it might be all plates produced by the process during a specified month. Changing the time period, production line, or operating conditions changes the population.

Definition 1.2 (Sample)

A **sample** is the subset of the population that is actually observed. ♣

The file used in this chapter contains a sample of $n = 180$ plates. A large sample is not automatically

representative: measurements taken only during one shift, for example, may systematically miss other operating conditions.

Definition 1.3 (Parameter and statistic)

A **parameter** is a numerical characteristic of a population. A **statistic** is a numerical characteristic calculated from a sample. ♣

We use Greek letters for population quantities and Latin letters for observed sample quantities. The population mean and standard deviation are denoted by μ and σ; their sample counterparts are $\bar{x}$ and s. A statistic describes the observed sample exactly, but it generally estimates an unknown population parameter.

Example 1.1Thickness study Suppose the population is all plates produced under the current process settings. Its mean thickness μ is a parameter. The mean of the 180 recorded measurements, $\bar{x} = 250.83\ \mu$m, is a statistic. The statistic is known once the file has been read; the population parameter remains unknown.

Observational units and variables

An **observational unit** is the item on which measurements are recorded. Each row in the plate dataset represents one plate. A **variable** is a recorded characteristic of that unit, represented by a column.

Definition 1.4 (Categorical and quantitative data)

Categorical data place observations into groups or categories. **Quantitative data** are numerical counts or measurements for which arithmetic has a meaningful interpretation. ♣

Quantitative variables are often divided into two types:

- A **discrete** variable takes countable values, such as the number of defects on a plate.
- A **continuous** variable is measured on a scale and may, in principle, take any value in an interval, such as thickness or response time.

Categorical variables may be **nominal**, with no inherent order, or **ordinal**, with an ordered set of categories. Production line is nominal; a severity rating such as low, medium, and high is ordinal. A numerical code does not turn a category into a quantitative measurement.

Variable	Role	Type	Meaning
`sample_id`	identifier	categorical label	identifies the measured plate
`line`	explanatory/grouping	nominal	production line A or B
`thickness_um`	response	continuous	measured thickness in μm

Table 1.1: Variables used in the Chapter 1 analysis. An identifier should not be averaged merely because it is stored as a number.

Before calculating anything, check the observational unit, column meanings, units, missing values, impossible values, duplicates, and how the sample was obtained. Statistical software cannot repair an ambiguous data-generating process.

1.3 Organising categorical and numerical data

For a categorical or discrete variable, a frequency table is usually a natural first summary.

Definition 1.5 (Frequency distribution)

A **frequency distribution** reports how often each category, value, or interval occurs. The **absolute frequency** is the count, the **relative frequency** is the count divided by the total number of observations, and the **cumulative relative frequency** is the running total in an ordered distribution.

♣

There are 90 observations from each production line, so each line has relative frequency $90/180 = 0.50$. Counts or bar charts are appropriate for production line. A histogram is not: the values A and B are categories, not intervals on a numerical scale.

For a continuous variable with many distinct values, we group observations into intervals called **bins**. A histogram places adjacent bars over the bins. The height represents a frequency, relative frequency, or density. Unlike a bar chart, the horizontal position and adjacency of histogram bars encode a numerical scale.

The appearance of a histogram depends on its bin boundaries and widths. Too few bins can hide structure; too many can exaggerate random fluctuation. Therefore, inspect more than one reasonable binning and do not treat a single histogram as an objective image of the population.

1.4 Measures of centre

No single number captures an entire distribution, but measures of centre help describe a typical observation.

Definition 1.6 (Sample mean)

For observations $x_1, x_2, \ldots, x_n$, the **sample mean** is

$$\bar{x} = \frac{1}{n} \sum_{i=1}^{n} x_i.$$

♣

The mean uses every observation and is the balance point of the data. Consequently, an unusually large or small value can move it substantially.

Definition 1.7 (Median)

After arranging the observations from smallest to largest, the **median** is the middle observation when n is odd and the mean of the two middle observations when n is even.

♣

The median is resistant to extreme observations: changing the largest value usually does not change which observations occupy the middle positions. It is often preferable to the mean for strongly skewed data, but reporting both is informative.

Definition 1.8 (Mode)

The **mode** is a value or category with the highest frequency. A dataset can have no unique mode or can have several modes.

♣

The mode is particularly useful for categorical data. For continuous measurements, exact repeated values depend on measurement precision, so a modal interval from a histogram is often more meaningful than an exact

mode.

Example 1.2Mean and median under an extreme value For the five values $8, 9, 10, 11, 12$, both the mean and median equal 10. If 12 is replaced by 52, the mean becomes 18 while the median remains 10. The mean reveals the effect of the extreme value; the median describes the centre of the majority. Neither value should be reported without context.

1.5 Measures of spread and position

Two processes can have the same mean and very different consistency. Measures of spread quantify this variation.

Definition 1.9 (Range)

The **range** is

$$x_{\max} - x_{\min}.$$

♣

The range is easy to interpret but depends entirely on two observations and usually grows as the sample size grows.

Definition 1.10 (Sample variance and sample standard deviation)

The **sample variance** and **sample standard deviation** are

$$s^2 = \frac{1}{n-1}\sum_{i=1}^{n}(x_i - \bar{x})^2, \qquad s = \sqrt{s^2}.$$

♣

Each deviation $x_i - \bar{x}$ measures displacement from the sample mean. Squaring makes negative and positive deviations contribute positively and gives large deviations greater weight. Variance is expressed in squared units; standard deviation returns to the original unit and is therefore easier to interpret.

Why $n - 1$?

Once $n - 1$ deviations from $\bar{x}$ are known, the final deviation is fixed because

$$\sum_{i=1}^{n}(x_i - \bar{x}) = 0.$$

Only $n - 1$ deviations are free to vary. Dividing the squared deviations by $n - 1$, rather than n, also makes s^2 an unbiased estimator of the population variance σ^2 under random sampling. This correction is important in later inference.

Pandas uses $n - 1$ for sample variance and standard deviation by default. NumPy uses n by default, so in this book we write `ddof=1` explicitly:

```
s_pandas = df["thickness_um"].std()
s_numpy = np.std(df["thickness_um"], ddof=1)
```

If a dataset is the entire population of interest and the aim is only to describe that finite population, division by n may be appropriate. State the convention and the population clearly.

Percentiles, quartiles, and IQR

Definition 1.11 (Percentile)

The p-th **percentile** is a value at or below which approximately $p\%$ of the observations fall.

Software packages use several interpolation conventions, which can produce slightly different percentiles in small samples. Record the convention when exact reproducibility matters.

Definition 1.12 (Quartiles and interquartile range)

The first quartile Q_1, median Q_2, and third quartile Q_3 are the 25th, 50th, and 75th percentiles. The **interquartile range** is

$$IQR = Q_3 - Q_1.$$

It measures the spread of the middle 50% of the observations.

Like the median, the IQR is resistant to extreme values. The pair median–IQR is therefore a useful summary for skewed data, while mean–standard deviation is most informative for reasonably symmetric data without influential extremes.

1.6 Visualising distributions and relationships

A numerical summary should normally be accompanied by a plot. Different plots reveal different features.

Histogram and distribution shape

A histogram can reveal centre, spread, gaps, clusters, and shape. Useful descriptive terms include:

- **symmetric**: the two sides are approximately mirror images;
- **right-skewed**: a longer tail extends toward larger values;
- **left-skewed**: a longer tail extends toward smaller values;
- **unimodal**: one prominent peak; and
- **multimodal**: two or more prominent peaks, possibly indicating mixed groups or operating states.

In many unimodal skewed datasets, the mean is pulled toward the longer tail. Thus, $\bar{x}$ is often larger than the median under right skew and smaller under left skew. This is a useful pattern, not a definition or universal rule.

Outlier flags and boxplots

An observation is not an error merely because it is unusual. It may be a recording mistake, a genuine rare event, a process change, or the most important observation in the dataset. A rule should flag values for investigation rather than delete them automatically.

Definition 1.13 ($1.5 \times IQR$ outlier rule)

Define the lower and upper fences by

$$L = Q_1 - 1.5\,IQR, \qquad U = Q_3 + 1.5\,IQR.$$

Observations below L or above U are flagged as potential outliers.

A boxplot displays the box from Q_1 to Q_3, a line at the median, and whiskers extending to the most extreme observations within the fences. Points outside the whiskers are drawn separately. Some software uses other whisker conventions, so check the documentation when exact interpretation matters.

Boxplots are excellent for comparing groups on a common scale. They do not show every feature: two distributions with different modality can have similar boxplots, and a short box does not imply that the process meets an engineering tolerance.

Scatterplots and ordered measurements

A scatterplot displays pairs (x_i, y_i) and can reveal direction, form, strength, groups, and unusual points in a relationship between two quantitative variables. Association does not by itself establish causation.

Plotting a measurement against collection order is also useful. In that case, order is not treated as an ordinary explanatory measurement; the plot is a diagnostic for drift, jumps, cycles, or changing variation. A stable histogram can conceal a change over time, which is why the collection order should be preserved whenever possible.

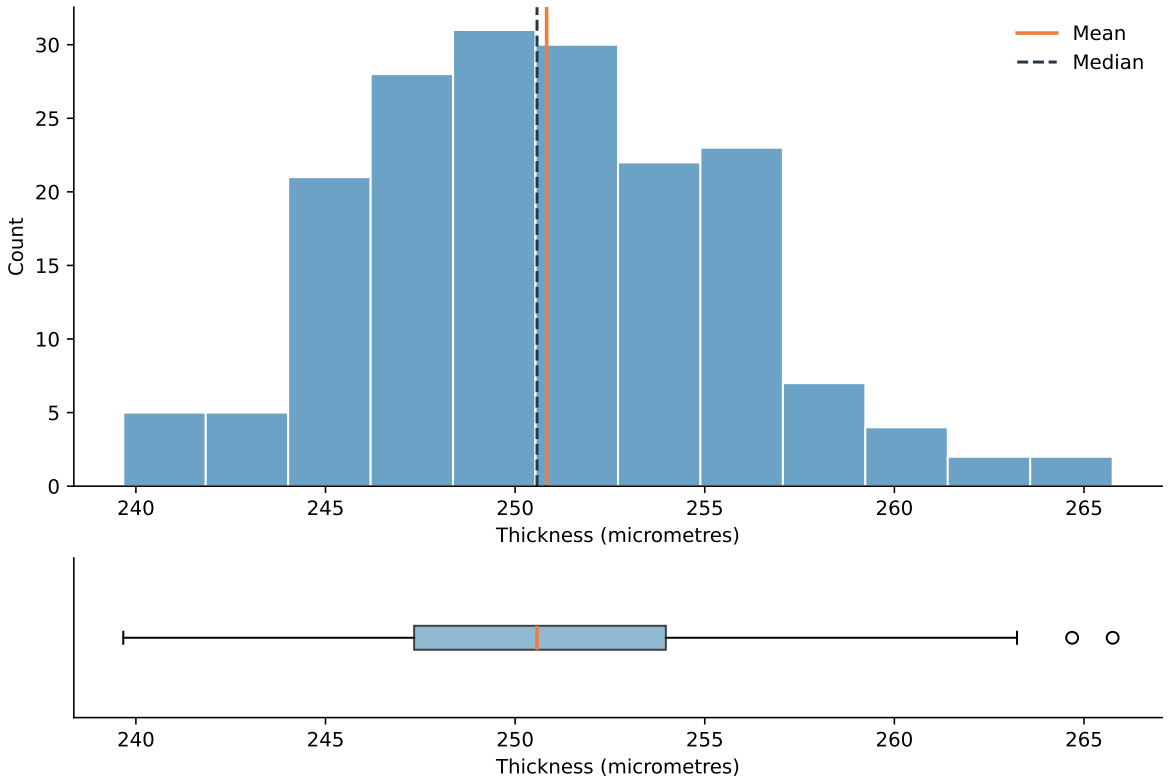


Figure 1.1: Distribution of the 180 plate-thickness measurements. The histogram shows overall shape, while the boxplot compactly displays quartiles and potential outliers.

1.7 Worked engineering example

The file `sensor_thickness.csv` contains 180 measurements in micrometres. The dataset is synthetic and reproducible, generated with NumPy's `default_rng(2026)` for teaching. Table 1.2 reports the principal summaries.

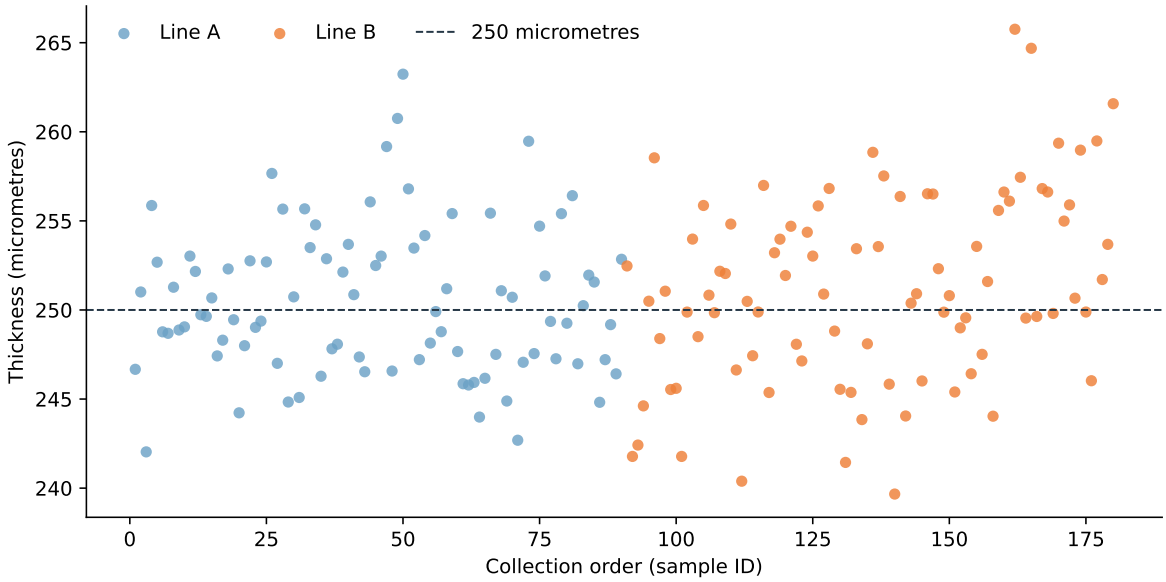


Figure 1.2: Thickness in collection order, coloured by production line. Ordered data can reveal structure that is hidden by a single aggregate distribution.

Statistic	**Thickness (μm)**
Count n	180
Mean $\bar{x}$	250.830
Median	250.577
Sample standard deviation s	4.809
Sample variance s^2	23.129
Minimum	239.668
Q_1	247.337
Q_3	253.974
Maximum	265.756
IQR	6.638

Table 1.2: Descriptive summary of all plate-thickness measurements. Values are computed from the distributed CSV file; rounding occurs only for display.

The IQR fences are

$$L = 247.33675 - 1.5(6.63750) = 237.3805,$$

$$U = 253.97425 + 1.5(6.63750) = 263.9305.$$

Two measurements, $265.756\,\mu$m and $264.687\,\mu$m, exceed the upper fence. They should be checked against measurement records and process conditions; the descriptive rule does not justify deleting them.

The mean is slightly above the median, and the upper flagged observations contribute to a modest right tail. The overall sample standard deviation is about $4.81\,\mu$m. In collection order, the later measurements on line B tend to be higher. This observation is a prompt for investigation, not yet evidence that one line or period has a different population mean. Such a claim requires a suitable design and inferential method.

This distinction is central:

> The sample shows what was observed. It does not, by itself, establish what will happen in the population or why the pattern occurred.

1.8 Python in practice

The following workflow is deliberately explicit. It keeps units, data checks, numerical summaries, plots, and interpretation together.

Import and audit

```python
from pathlib import Path

import matplotlib.pyplot as plt
import numpy as np
import pandas as pd

DATA = Path("data")
df = pd.read_csv(DATA / "sensor_thickness.csv")

print(df.head())
print(df.info())
print(df.isna().sum())
print(df.duplicated(subset="sample_id").sum())
```

For this file, each row is one plate, `sample_id` is unique, and `thickness_um` is the analysis variable. Keep the unit μm in plot labels and reported conclusions.

Compute summaries explicitly

```python
x = df["thickness_um"]

summary = {
    "n": x.count(),
    "mean": x.mean(),
```

```
    "median": x.median(),
    "sample_variance": x.var(ddof=1),
    "sample_sd": x.std(ddof=1),
    "minimum": x.min(),
    "maximum": x.max(),
}

q1, q3 = x.quantile([0.25, 0.75])
iqr = q3 - q1
lower = q1 - 1.5 * iqr
upper = q3 + 1.5 * iqr
flagged = df.loc[(x < lower) | (x > upper)]

print(pd.Series(summary))
print(f"Q1={q1:.3f}, Q3={q3:.3f}, IQR={iqr:.3f}")
print(flagged[["sample_id", "line", "thickness_um"]])
```

The call `x.describe()` is a convenient check, but explicit calculations make the conventions visible. In particular, `ddof=1` documents that s, not the population standard deviation, is intended.

Plot the distribution and collection order

```
fig, axes = plt.subplots(2, 1, figsize=(8, 6),
                         gridspec_kw={"height_ratios": [3, 1]})

axes[0].hist(x, bins="fd", color="#6ca2c6", edgecolor="white")
axes[0].set(xlabel="Thickness (micrometres)", ylabel="Count")

axes[1].boxplot(x, vert=False)
axes[1].set_xlabel("Thickness (micrometres)")

fig.tight_layout()
plt.show()
```

```
fig, ax = plt.subplots(figsize=(8, 4))
for line, group in df.groupby("line"):
    ax.scatter(group["sample_id"], group["thickness_um"],
               label=f"Line {line}", s=24, alpha=0.8)

ax.set(xlabel="Collection order (sample ID)",
       ylabel="Thickness (micrometres)")
ax.legend()
fig.tight_layout()
plt.show()
```

End the analysis in ordinary language. For example:

> The 180 recorded plates have a mean thickness of $250.83\,\mu$m and a sample standard deviation of $4.81\,\mu$m. The distribution is broadly concentrated around $250\,\mu$m, but the IQR rule flags two high

measurements and the ordered plot suggests higher later values on line B. The measurement records and sampling process should be checked before drawing a process-wide conclusion.

1.9 Common pitfalls and chapter summary

Common pitfalls

- **Confusing population and sample.** Describe precisely which units the conclusion concerns.
- **Treating identifiers as measurements.** A numerical sample ID is still a label.
- **Reporting only the mean.** Pair centre with spread and a plot.
- **Using NumPy's default variance unintentionally.** Set `ddof=1` for s^2 and s.
- **Calling every flagged value an error.** Investigate before correcting or excluding it.
- **Ignoring collection order or groups.** Aggregation can hide process changes.
- **Inferring from a graph alone.** A visible sample difference is descriptive evidence, not a completed hypothesis test.
- **Omitting units.** A value such as 4.81 is incomplete without μm.

Chapter summary

Descriptive statistics organises and communicates observed data. The mean, median, and mode describe centre; the range, variance, standard deviation, and IQR describe spread; and quartiles describe position. Histograms show distribution shape, boxplots compactly show centre, spread, and flagged observations, and scatterplots or ordered plots reveal relationships and process structure. These tools help formulate better questions, but they do not replace probability models, sampling arguments, or statistical inference.

1.10 Exercises

1. For each variable below, identify the observational unit and classify the variable as nominal, ordinal, discrete quantitative, or continuous quantitative:
   (a). component lifetime in hours;
   (b). defect severity (minor, major, critical);
   (c). number of failed tests in a build;
   (d). server region; and
   (e). request latency in milliseconds.
2. A team records response times $42, 45, 47, 48, 53, 55, 110$ ms.
   (a). Compute the mean and median.
   (b). Explain which measure better describes a typical request and what information the other measure retains.
   (c). Compute the range and sample standard deviation using Python with `ddof=1`.
3. Use `sensor_thickness.csv`.
   (a). Reproduce Table 1.2 without rounding intermediate calculations.
   (b). Verify that Pandas and NumPy agree on the sample standard deviation when `ddof=1` is used.
   (c). Compare the result with NumPy's default. Explain the difference.

4. Construct histograms with 5, 10, 20, and Freedman–Diaconis bins. Which features are stable across the choices, and which depend on the binning? Include units on both axes where relevant.
5. Compute the IQR fences and list the flagged plate measurements. For each plausible cause below—recording error, rare valid plate, calibration problem, or process shift—state what additional evidence you would seek.
6. Produce side-by-side boxplots of thickness for lines A and B, together with a table containing n, mean, median, and sample standard deviation for each line. Write a descriptive conclusion that does not make an inferential claim.
7. Plot thickness against collection order. Compare what this plot reveals with what is visible in the aggregate histogram. Why should collection order normally be retained?
8. Create a deliberately misleading plot of the thickness data by removing units or choosing a distorted axis. Then correct it and explain why the revised version is more responsible.
9. Write a short data-quality note for the dataset. Address provenance, synthetic-data status, missingness, duplicates, units, sampling limitations, and any variables that should not be treated as ordinary predictors in an initial analysis.
10. An engineer says: "Line B has a larger sample mean, so line B produces thicker plates." Explain precisely what the data establishes and what would be required to support the process-wide claim.

Chapter 2 Probability for Data Analysis

Data describe what has been observed. Probability describes uncertainty about what may be observed next and provides the language needed to connect a sample to a wider process. In engineering, that uncertainty may concern a component failure, a packet loss, a false alarm, or the unknown cause of a detected fault.

This chapter follows the notation and core definitions in *Mathematics for Software Engineering*, but selects only the material needed for practical data analysis. We use events, tables, and probability trees rather than a catalogue of counting formulas. The binomial coefficient is introduced in Chapter 3, where it is needed.

2.1 Engineering question and learning outcomes

Our running example is an automated optical inspection system that flags possible solder defects on circuit boards. We ask:

> When the inspection system flags a board, how strongly should that result change our belief that the board is actually defective?

After studying this chapter, you should be able to:

- represent a random experiment using a sample space and events;
- translate not, and, and or into complements, intersections, and unions;
- use complement and addition rules without double-counting;
- calculate conditional probabilities from a contingency table;
- distinguish mutually exclusive events from independent events;
- use multiplication rules and probability trees;
- apply the law of total probability and Bayes' theorem; and
- distinguish sensitivity, false-positive rate, and positive predictive value.

2.2 Random experiments, sample spaces, and events

Definition 2.1 (Random experiment)

A **random experiment** is a repeatable process whose outcome is not known with certainty before it is observed. ♣

Random does not mean unstructured. Repeated trials under apparently similar conditions can produce different outcomes, and this uncertainty is part of the model.

Definition 2.2 (Sample space, outcome, and event)

The **sample space**, denoted by S, is the set of all possible outcomes of a random experiment. An **outcome** is one element of S. An **event** is a subset of S. ♣

The sample space depends on the question. For one board, it could be

$$S = \{\text{flagged}, \text{not flagged}\}.$$

If the physical condition and test result are both relevant, use

$$S = \{(\text{defective}, \text{flagged}), (\text{defective}, \text{not flagged}),$$
$$(\text{good}, \text{flagged}), (\text{good}, \text{not flagged})\}.$$

This representation makes false positives and false negatives visible.

Let D denote a defective board and F a flag:

- D^c: the board is not defective;
- $D \cap F$: the board is defective and flagged;
- $D \cup F$: the board is defective or flagged, including both; and
- $D - F = D \cap F^c$: the board is defective but not flagged.

In probability, or is inclusive unless but not both is stated.

Probability as a model

A probability may come from a physical model, long-run relative frequencies, historical data, or explicit engineering judgement. Whatever its origin, it must satisfy the following rules.

Definition 2.3 (Probability rules)

A probability model satisfies:

1. $0 \le P(A) \le 1$ for every event A;
2. $P(S) = 1$; and
3. if $A_1, A_2, \ldots$ are mutually exclusive, then
$$P\left(\bigcup_i A_i\right) = \sum_i P(A_i).$$



An empirical failure rate is not automatically a timeless law. A rate estimated under one operating condition may be inappropriate after a process change.

2.3 Complements, unions, and intersections

Theorem 2.1 (Complement and addition rules)

For events A and B,

$$P(A^c) = 1 - P(A)$$

and

$$P(A \cup B) = P(A) + P(B) - P(A \cap B).$$

If A and B are mutually exclusive, $P(A \cap B) = 0$.



The subtraction prevents the intersection from being counted twice.

Example 2.1Two inspection warnings Suppose 12% of boards trigger a solder warning, 7% trigger an alignment warning, and 3% trigger both. If S_o and A_l denote the events, then

$$P(S_o \cup A_l) = 0.12 + 0.07 - 0.03 = 0.16.$$

Thus, 16% trigger at least one warning.

The complement rule is often easiest for at least one. If three independent backup components each fail with probability 0.02, the probability that at least one fails is

$$1 - (1 - 0.02)^3 \approx 0.0588.$$

The multiplication requires independence.

2.4 Conditional probability and contingency tables

New information changes the reference group. If a board is known to be flagged, compare it only with flagged boards.

Definition 2.4 (Conditional probability)

For events A and B with $P(B) > 0$,

$$P(A \mid B) = \frac{P(A \cap B)}{P(B)}.$$

♣

The condition is to the right of the vertical bar. In general,

$$P(F \mid D) \neq P(D \mid F).$$

The first asks how the test behaves among defective boards; the second asks how many flagged boards are actually defective.

Natural frequencies

Suppose historical information gives

$$P(D) = 0.02, \qquad P(F \mid D) = 0.95, \qquad P(F \mid D^c) = 0.08.$$

For a hypothetical batch of 10,000 boards, the expected counts are:

	Flagged	**Not flagged**	**Total**
Defective	190	10	200
Good	784	9016	9800
Total	974	9026	10000

Table 2.1: Natural-frequency representation of the inspection model. Counts are expected counts, not newly observed data.

The table gives

$$P(F \mid D) = \frac{190}{200} = 0.95,$$
$$P(F \mid D^c) = \frac{784}{9800} = 0.08,$$
$$P(D \mid F) = \frac{190}{974} \approx 0.195.$$

These are sensitivity, false-positive rate, and positive predictive value, respectively. A sensitivity of 95% does not imply that 95% of flagged boards are defective.

2.5 Multiplication rules and probability trees

Theorem 2.2 (Multiplication rule)

For events A and B,

$$P(A \cap B) = P(A)P(B \mid A) = P(B)P(A \mid B),$$

whenever the stated conditional probabilities are defined.

A probability tree starts with a base-rate branch and continues with conditional branches. Multiply along a path and add mutually exclusive paths.

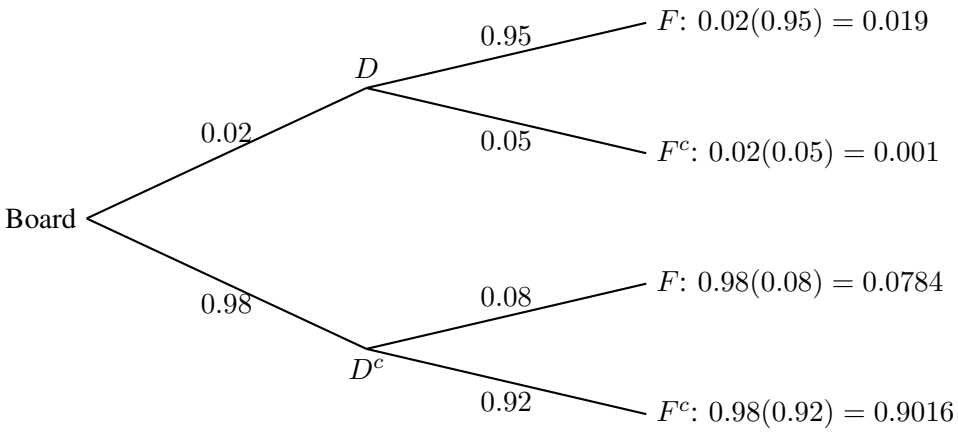


Figure 2.1: Probability tree for the optical inspection system.

2.6 Independence and mutual exclusivity

Definition 2.5 (Independence)

Events A and B are **independent** if

$$P(A \cap B) = P(A)P(B).$$

When $P(B) > 0$, this is equivalent to $P(A \mid B) = P(A)$.

Independence means that learning whether one event occurred does not change the probability of the other. It must be supported by the process or model, not assumed for convenience.

Definition 2.6 (Mutually exclusive events)

Events A and B are **mutually exclusive** if $A \cap B = \emptyset$.

If mutually exclusive events both have positive probability, observing one rules out the other, so they are dependent. For example, F and F^c are mutually exclusive but not independent when $0 < P(F) < 1$. Results from consecutive boards should be modelled as independent only if common machine drift or batch effects are negligible.

2.7 Total probability and Bayes' theorem

A set of mutually exclusive events whose union is S is a **partition**. Events D and D^c partition the board population.

Theorem 2.3 (Law of total probability)

If $B_1, \dots, B_k$ form a partition of S, then

$$P(A) = \sum_{i=1}^{k} P(A \mid B_i) P(B_i).$$

♡

For the inspection system,

$$\begin{aligned} P(F) &= P(F \mid D)P(D) + P(F \mid D^c)P(D^c) \\ &= 0.95(0.02) + 0.08(0.98) = 0.0974. \end{aligned}$$

Theorem 2.4 (Bayes' theorem)

If $B_1, \dots, B_k$ form a partition and $P(A) > 0$, then

$$P(B_j \mid A) = \frac{P(A \mid B_j)P(B_j)}{\sum_{i=1}^{k} P(A \mid B_i)P(B_i)}.$$

For a two-part partition,

$$P(B \mid A) = \frac{P(A \mid B)P(B)}{P(A \mid B)P(B) + P(A \mid B^c)P(B^c)}.$$

♡

Bayes' theorem reverses the direction of conditioning. The numerator is the joint probability of the hypothesis and evidence; the denominator is the total probability of the evidence across every path.

2.8 Worked engineering example

$$\begin{aligned} P(D \mid F) &= \frac{P(F \mid D)P(D)}{P(F \mid D)P(D) + P(F \mid D^c)P(D^c)} \\ &= \frac{0.95(0.02)}{0.95(0.02) + 0.08(0.98)} \approx 0.195. \end{aligned}$$

A flagged board has a 19.5% probability of being defective under this model. About 80.5% of flags are false positives. This does not contradict the 95% sensitivity: 190 of 200 defective boards are caught, but the much larger group of good boards produces 784 false alarms.

A flag still increases risk from the 2% base rate to 19.5%. Whether that is sufficient to scrap a board depends on the costs of confirmation, missed defects, and unnecessary scrap. A sensible workflow may send flagged boards to a more accurate confirmatory test.

The same inspection system produces different predictive values in populations with different defect rates.

2.9 Python in practice

Figure 2.2 visualises the base-rate effect before we reproduce the calculation in Python.

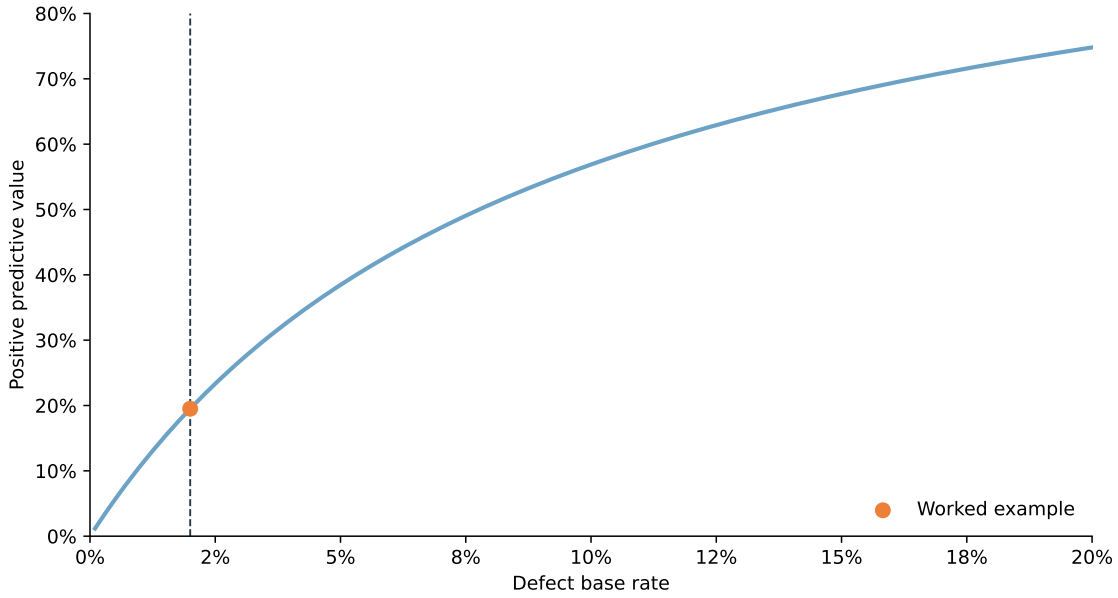


Figure 2.2: Positive predictive value as a function of defect base rate for sensitivity 0.95 and false-positive rate 0.08.

```python
import numpy as np
import pandas as pd
import matplotlib.pyplot as plt

p_defect = 0.02
sensitivity = 0.95
false_positive_rate = 0.08

p_flag = (
    sensitivity * p_defect
    + false_positive_rate * (1 - p_defect)
)
ppv = sensitivity * p_defect / p_flag

print(f"P(flag) = {p_flag:.4f}")
print(f"P(defect | flag) = {ppv:.4f}")
```

Natural frequencies make the denominators visible:

```python
n = 10_000
n_defective = round(n * p_defect)
n_good = n - n_defective

tp = round(n_defective * sensitivity)
fn = n_defective - tp
fp = round(n_good * false_positive_rate)
```

```
tn = n_good - fp

table = pd.DataFrame(
    {"flagged": [tp, fp], "not_flagged": [fn, tn]},
    index=["defective", "good"],
)
table["total"] = table.sum(axis=1)
table.loc["total"] = table.sum(axis=0)
print(table)
```

Simulation checks the long-run interpretation, but does not replace the exact calculation:

```
rng = np.random.default_rng(2026)
n_sim = 100_000

defective = rng.random(n_sim) < p_defect
flagged = np.where(
    defective,
    rng.random(n_sim) < sensitivity,
    rng.random(n_sim) < false_positive_rate,
)

simulated_ppv = defective[flagged].mean()
print(f"Simulated P(defect | flag) = {simulated_ppv:.4f}")
```

Under a 2% defect base rate, 95% sensitivity, and an 8% false-positive rate, a flagged board has a 19.5% probability of being defective. A flag should trigger confirmation rather than be treated as proof.

2.10 Common pitfalls and chapter summary

Common pitfalls

- **Reversing the condition.** $P(F \mid D)$ and $P(D \mid F)$ answer different questions.
- **Ignoring the base rate.** Sensitivity alone cannot determine positive predictive value.
- **Double-counting a union.** Subtract $P(A \cap B)$ unless the events are mutually exclusive.
- **Confusing exclusivity with independence.** Disjoint positive-probability events are dependent.
- **Multiplying marginal probabilities automatically.** $P(A \cap B) = P(A)P(B)$ requires independence.
- **Treating inputs as timeless facts.** Base rates and error rates can change across populations and conditions.

Chapter summary

Events are subsets of a sample space. Complements handle not, intersections handle and, and unions handle inclusive or. Conditional probability changes the reference group. Independence means conditioning does not change a probability; it is not mutual exclusivity. Probability trees and the multiplication rule construct joint probabilities, total probability adds mutually exclusive paths, and Bayes' theorem reverses conditioning.

2.11 Exercises

1. A server request can succeed, time out, or return an application error. Define a sample space and the event that the request fails. Explain how the sample space changes if response time is recorded continuously.
2. Let $P(A) = 0.18$, $P(B) = 0.25$, and $P(A \cap B) = 0.09$ for memory and latency warnings. Find $P(A \cup B)$, $P(A - B)$, and the probability of neither warning.
3. A quality sample contains 72 flagged defective items, 8 unflagged defective items, 45 flagged good items, and 875 unflagged good items. Compute sensitivity, false-positive rate, positive predictive value, negative predictive value, and accuracy. Interpret each denominator.
4. Give one engineering example of mutually exclusive events and one of events that could reasonably be independent. Justify both.
5. A component comes from line A with probability 0.65 and line B with probability 0.35. Failure probabilities are 0.01 and 0.04. Draw a tree, find the overall failure probability, and find the probability that a failed component came from line B.
6. Reproduce Table 2.1 in Pandas and verify the three displayed conditional probabilities from its row and column totals.
7. Reproduce Figure 2.2. Compare predictive value at base rates 0.5%, 2%, 10%, and 20%. Explain why sensitivity alone is insufficient.
8. Simulate the inspection example for $n = 1000$, 10 000, and 100 000 with `default_rng(2026)`. Compare with the exact result.
9. Reduce the false-positive rate from 0.08 to 0.01 while sensitivity remains 0.95. Calculate the new predictive value and expected false alarms per 10,000 boards.
10. Correct the claim that a detector with 95% sensitivity gives every flagged board a 95% probability of being defective.

Chapter 3 Random Variables and Discrete Distributions

3.1 Engineering question and learning outcomes

How can counts and binary outcomes be modelled probabilistically?

3.2 Random variables, PMFs, and CDFs

3.3 Expectation and variance

3.4 Bernoulli and binomial models

3.5 Poisson models

3.6 Python in practice

3.7 Common pitfalls and chapter summary

3.8 Exercises

Chapter 4 Continuous Distributions in Practice

4.1 Engineering question and learning outcomes

How can continuous measurements, waiting times, and lifetimes be modelled?

4.2 Densities, distribution functions, and quantiles

4.3 Uniform, normal, and exponential models

4.4 Model checking and QQ plots

4.5 Worked engineering example

4.6 Python in practice

4.7 Common pitfalls and chapter summary

4.8 Exercises

Part II

Sampling and Statistical Inference

Chapter 5 Sampling Distributions and the Central Limit Theorem

5.1 Engineering question and learning outcomes

Why do estimates vary from sample to sample, and what controls that variation?

5.2 Statistics as random variables

5.3 The sampling distribution of the mean

5.4 Standard error

5.5 The central limit theorem

5.6 Python simulation

5.7 Common pitfalls and chapter summary

5.8 Exercises

Chapter 6 Estimation and Confidence Intervals

6.1 Engineering question and learning outcomes

How precisely has a population quantity been estimated from a sample?

6.2 Estimators, estimates, bias, and precision

6.3 Intervals for a mean

6.4 Intervals for a proportion

6.5 Coverage and interpretation

6.6 Python in practice

6.7 Common pitfalls and chapter summary

6.8 Exercises

Chapter 7 Hypothesis Testing I: Principles and One-Sample Tests

7.1 Engineering question and learning outcomes

Does a process provide evidence against a specified target or baseline?

7.2 Hypotheses, significance levels, and p-values

7.3 Type I and type II errors

7.4 One-sample tests for a mean and proportion

7.5 Effect size and practical importance

7.6 Python in practice

7.7 Common pitfalls and chapter summary

7.8 Exercises

Chapter 8 Hypothesis Testing II: Comparing Two Groups

8.1 Engineering question and learning outcomes

Do two processes, systems, or treatments differ in a practically meaningful way?

8.2 Independent and paired designs

8.3 Welch's two-sample t-test

8.4 The paired t-test

8.5 Comparing two proportions

8.6 Python in practice

8.7 Common pitfalls and chapter summary

8.8 Exercises

Chapter 9 Analysis of Variance

9.1 Engineering question and learning outcomes

How can several group means be compared without inflating the false-positive rate?

9.2 The one-way ANOVA model

9.3 Between-group and within-group variation

9.4 The F-test and its assumptions

9.5 Post-hoc comparisons

9.6 Python in practice

9.7 Common pitfalls and chapter summary

9.8 Exercises

Part III

Modelling and Integrated Analysis

Chapter 10 Simple Linear Regression

10.1 Engineering question and learning outcomes

How can a linear relationship between two quantitative variables be modelled and interpreted?

10.2 The regression model and least squares

10.3 Correlation and coefficient of determination

10.4 Inference for the slope

10.5 Residual diagnostics and prediction

10.6 Python in practice

10.7 Common pitfalls and chapter summary

10.8 Exercises

Chapter 11 Categorical Data and Chi-Square Tests

11.1 Engineering question and learning outcomes

How can observed category counts be compared with a model or across groups?

11.2 Counts, proportions, and expected counts

11.3 Chi-square goodness-of-fit tests

11.4 Chi-square tests of independence

11.5 Residuals and follow-up interpretation

11.6 Python in practice

11.7 Common pitfalls and chapter summary

11.8 Exercises

Chapter 12 Integrated Statistical Analysis and Exam Preparation

12.1 Engineering question and learning outcomes

How can a defensible analysis be carried from raw data to an engineering recommendation?

12.2 From question to data audit

12.3 Method selection

12.4 Assumptions, uncertainty, and limitations

12.5 Communicating a recommendation

12.6 Integrated Python case study

12.7 Common pitfalls and chapter summary

12.8 Exercises

References

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Appendix A: Python and Data Setup

This appendix supports the preliminary STA1 setup session. It will cover the course environment, reproducible random-number generation, data paths, DataFrame inspection, plotting conventions, and notebook reporting workflow.

A.1 Recommended environment

A.2 Core packages

A.3 Reproducible analysis

A.4 Loading course data

A.5 Notebook and reporting workflow